ROOTFLASH TECHNOLOGY | CANDIDATE ASSESSMENT

# FIRST-ROUND TECHNICAL ASSIGNMENT

Solar CRM Management System

**MANDATORY BUILD RULE**

**Build the project using Claude AI only. No manual coding. Maximum total time: 30 hours.**

## Objective

Build a working, demonstrable Solar CRM web application covering the customer journey from lead capture and site survey through hybrid solar system sizing, quotation, order/project execution, inventory, reporting, after-sales service, and dynamically configurable system settings. This is a time-boxed prototype challenge: prioritize connected workflows, believable data, correct calculations, configuration-driven behavior, and clear technical reasoning.

| Item | Requirement |
|---|---|
| **Time limit** | 30 hours maximum from assignment receipt |
| **Required AI** | Claude AI only for generating, editing, or debugging project code |
| **Submission** | Runnable app + source repository + README + Claude AI evidence + full-project screen recording |
| **Next round** | Technical interview based on the actual implementation and recorded walkthrough |

Important: this assignment is for evaluation only. Build an original implementation. Do not copy proprietary products, private repositories, or third-party source code.

# 1. AI-only development rules and submission checklist

## Mandatory rules

- Claude AI must be the only AI coding assistant used to generate, modify, refactor, or debug project code. Do not use other AI coding agents or code generators.
- No manual coding: do not hand-write or directly edit application source code. Use Claude to generate or revise code, then run and test it. Normal IDE/terminal operation, environment setup, database tools, Git operations, and UI testing are allowed.
- Maximum duration is 30 hours total beginning when the assignment is received. Stop at the deadline and submit the current state. Do not present incomplete or simulated features as fully functional.
- Keep Claude prompts and responses that relate to architecture, code generation, debugging, and testing. Provide relevant exports or screenshots as evidence.
- Use fictional seed data. Do not place real customer data, passwords, API tokens, or company secrets in the repository. Store credentials in environment variables or a secure server-side secret mechanism.
- You must understand the submitted code. AI-generated code that you cannot explain may count against the assessment.

## Required deliverables

- Runnable, responsive web application with connected core workflows and demo credentials.
- Source repository/archive, database migrations/schema, seed data, dependency/lock files and a safe environment sample file.
- README with setup, stack, architecture, database setup, demo roles, test steps, deployment notes, and known limitations.
- Full-project screen-recording video showing the live running app, every major module, key workflows, dynamic settings, WhatsApp test/demo behavior, calculator inputs and outputs, exports, and any limitations.
- Claude AI evidence: relevant prompt history/export or screenshots plus a short note describing validation and debugging.
- Feature status sheet marking each required feature as Fully working, Simulated, Partial, or Not implemented.

## Suggested screen-recording structure (12-20 minutes)

| Time | What to demonstrate |
|---|---|
| **1-2 min** | Overview, stack, architecture and database relationships. |
| **4-6 min** | Lead creation/assignment, customer visit, hybrid calculator, branded quote PDF and quote-to-project conversion. |
| **3-5 min** | Project milestones, inventory, tickets, dashboard, reports and user permissions. |
| **3-5 min** | Settings tab: change app name/logo, change password, edit dropdown/workflow/calculator settings, configure WhatsApp test mode, demonstrate changes taking effect; finish with limitations and Claude evidence. |

# 2. Product brief and end-to-end workflow

Create a centralized CRM for a solar EPC/installation business. It should connect sales, field survey, hybrid system sizing, quotations, execution, inventory, service, analytics and a broad dynamic settings center.

**PRIMARY WORKFLOW**

**Lead -> follow-up/qualification -> customer/site visit -> hybrid solar calculation -> quotation/proposal -> accepted quotation -> sales order/project -> materials/team allocation -> installation milestones -> commissioning/handover -> invoice/payment visibility -> service ticket/AMC.**

## Minimum quality expectations

- Use consistent navigation, clear page titles, search/filter controls, validation messages, useful empty/loading/error states and responsive forms.
- Use linked records rather than duplicate disconnected entries: a lead/customer should link to visits, calculations, quotations, orders/projects, documents and tickets.
- Use unique reference IDs for leads, visits, quotations, orders/projects, products, invoices, tickets and AMC contracts.
- Use stored data to calculate dashboard metrics and reports. Do not use decorative hard-coded KPIs.
- Dynamic business configuration belongs in the Settings tab and database. Changing a setting must affect the relevant UI, calculations, templates, exports or workflows without code edits or redeployment, where technically applicable.
- If an external service such as WhatsApp, email, GPS/maps, e-signature, live subsidy data or payment gateway is not integrated, label it as Demo/Simulated and document production setup.

## Core demo user roles

| Role | Expected access |
|---|---|
| **Super Admin** | All modules, settings, users, roles/permissions, company configuration, audit logs and integrations. |
| **Sales Manager** | Team leads, assignment, approvals, quotes, sales pipeline and team reports. |
| **Sales Executive** | Assigned leads, follow-ups, visits, customers, calculators, quotations and own pipeline. |
| **Surveyor / Engineer** | Assigned site visits, survey forms, GPS/site evidence and technical notes. |
| **Project Manager / Operations** | Accepted orders, projects, milestones, materials, team allocation and installation completion. |
| **Inventory / Finance / Service** | Relevant stock, procurement, invoice/payment visibility, support tickets and AMC functions. |

# 3. Required modules and field-level details

## A. Dashboard and activity center

- Summary cards: new leads, follow-ups due/overdue, scheduled visits, quotations drafted/sent/accepted, active projects, open tickets, recorded revenue and outstanding amounts when payment data exists.
- Charts/lists: lead-stage pipeline, source mix, sales by team member, quote conversion, monthly revenue, project status, recent activity, upcoming tasks and notifications.
- Filters: date range, branch, team member, lead source and status. KPI cards should open the underlying filtered records.
- Activity center: timestamped events, task reminders, overdue highlights, read/unread notifications and action history.

## B. Lead management

| Field group | Fields and behavior |
|---|---|
| **Identity** | Auto lead ID, created date/time, lead title, customer type, full name/company, contact person, mobile, alternate phone, email. |
| **Location** | Address, city, district, state, PIN/postal code, GPS coordinates if available, branch/territory. |
| **Solar need** | System type (hybrid, solar water pump, EV charging or other supported project type), property segment, roof/site type, average monthly bill/units, sanctioned load, desired capacity, priority and budget range. |
| **Sales tracking** | Source, campaign/source details, assigned owner, stage, loss reason, probability, expected value, notes and tags. |
| **Follow-up/files** | Next follow-up, reminder, outcome, call/meeting notes, documents/photos, interaction history, created/updated by and timestamps. |
| **Actions** | Add/edit/view, assign/reassign, import/export where feasible, search/filter/sort, duplicate check, follow-up scheduling and customer conversion. |

## C. Customer visit and site inspection

- Visit ID, linked lead/customer, schedule, assigned surveyor, status, site address, GPS coordinates, reschedule/cancellation reason and visit history.
- Survey fields: property type, roof type/material, usable area, orientation/tilt, shading/obstructions, structural concerns, photos, electrical room/meter photos, site access and safety notes.
- Utility details: bill/consumption evidence, monthly units, sanctioned load, single/three-phase, utility/DISCOM, meter/connection type, proposed hybrid system and tentative capacity.
- Visit outcome: measurements, feasibility, customer needs, objections, feedback, recommended next step, preliminary bill of materials and downloadable visit report.
- Map/GPS is a real integration only when configured; otherwise allow manual coordinates and label the map preview as demo.

## D. Tasks, calendar and follow-ups

- Tasks can be linked to leads, customers, quotes, visits, projects and tickets. Fields: title, description, owner, due date/time, priority, status, reminder and outcome.
- Provide list/calendar view, overdue and due-today views, completion timestamps and linked-record activity history.

# 4. Solar calculators and calculation rules

The calculator is a key evaluation area. It must not return arbitrary or fixed results. Show inputs, formulas/assumptions, intermediate totals, units, final estimate and a printable/exportable report. Results are indicative estimates, not an engineering certification.

## A. Hybrid solar calculator (replaces separate off-grid and on-grid calculator requirements)

- Load table: appliance/load name, quantity, rated watts, hours/day, usage pattern, critical/non-critical classification and optional starting/surge factor. Calculate connected load, daily energy (Wh/kWh) and peak simultaneous load.
- Solar and utility inputs: average monthly units and/or bill, tariff per unit, usable roof area, proposed PV capacity, panel wattage, peak sun hours or generation assumption, performance ratio/system loss factor, self-consumption assumption, grid availability and export/net-metering assumption where applicable.
- Battery/inverter inputs: backup duration target, system voltage, battery chemistry, nominal battery capacity, usable depth of discharge, battery/inverter efficiency, inverter continuous and surge rating, charge-controller/inverter limits and optional generator input.
- Financial inputs: panel/inverter/battery/BOS prices, GST rate, installation, transport, structure/civil work, wiring/protection, engineering, other charges, O&M, and editable subsidies/export rates where applicable.
- Outputs: recommended PV kWp and panel count, roof-area-constrained system size if enabled, daily/annual energy estimate, battery nominal and usable kWh/Ah requirement, inverter rating, estimated backup time, expected solar contribution, grid/generator support assumptions, self-consumed/exported units if configured, cost breakdown, total estimate and PDF report.
- Display each selected input, unit and assumption beside results. Do not apply one universal battery/inverter rule without showing the inputs and calculation path.
- If real tariff, net metering or subsidy rules are used, show applicable state/utility, source/reference and effective date. Otherwise mark all such values Sample/Demo and make them editable in Settings.
- Use configurable formulas and documented assumptions. Validate physical plausibility, non-negative values, units, efficiency ranges and zero denominators. State when detailed engineering validation is required.

## B. Solar investment / income calculator

- Inputs: upfront cost, applicable grant/subsidy, optional loan amount/interest/tenure, estimated generation, self-consumption, export price, utility tariff, degradation, tariff escalation, annual maintenance and other recurring costs.
- Outputs: net initial investment, monthly/annual savings or revenue, annual operating cost, net cash flow, payback period, ROI for a selected period and an assumptions report. NPV/IRR are bonus items.

## C. EV charging income calculator

- Inputs: charger type/power (kW), number of charging points, sessions/day, average kWh/session, operating days/month, selling price, electricity purchase cost, utilization, rent/staff/maintenance/other monthly costs and optional capital cost.
- Outputs: energy sold, gross monthly/annual revenue, electricity cost, operating expenses, estimated net profit, utilization assumptions, break-even and payback estimate when capital cost is provided.

## Calculator acceptance tests

| Test | Expected behavior |
|---|---|
| **Hybrid sizing** | Changing appliance quantity/hours, backup target, panel wattage, peak sun hours, battery usable depth or efficiency changes relevant outputs logically. |
| **Generation/savings** | Changing tariff, generation/loss assumption, self-consumption or export rate changes projected savings and payback. |
| **Cost estimate** | Changing product prices, GST, installation or transport updates line items and total; generated PDF matches on-screen results. |
| **Validation** | Reject negative inputs, implausible efficiencies, invalid denominators and non-numeric inputs; show understandable errors. |

# 5. Quotations, product catalogue and customer documents

## A. Quotation and proposal management

- Common fields: auto quotation number, revision, created/valid-until dates, customer/lead/site, owner, system type/capacity, status (Draft, Review, Sent, Accepted, Rejected, Expired), internal notes, customer terms and payment schedule.
- Itemized lines: product/service, SKU, description/specification, quantity, unit, unit price, discount, taxable value, GST rate/tax, line total and stock availability. Support installation, civil work, transport, accessories, AMC and other charges.
- Totals: subtotal, discount, taxable value, configured CGST/SGST or IGST treatment, round-off if used, grand total, advance and balance due. Tax rates/rules are configurable; do not hard-code universal tax advice.
- Hybrid proposal: panel/inverter/battery/structure/protection/accessories, load and backup assumptions, scope, exclusions, commissioning, warranty, maintenance and validity.
- Solar water-pump proposal: pump model/capacity, head/discharge/water requirement, controller, panels, structure, accessories, installation and configured subsidy assumptions.
- Output/share: branded PDF, print view, QR code to quote verification/detail, digital signature area, activity trail, and Email/WhatsApp share action. Do not say a message was sent unless the provider confirms it; demo actions must be labeled.

## B. Product catalogue / product master

- Product ID/SKU, category, brand/model/title, description, technical specs, wattage/capacity, unit, purchase cost, sale price, GST rate, warranty, vendor, status, image, serial/batch tracking, available stock and reorder threshold.
- Categories should cover solar panels, hybrid inverters, batteries, pumps, chargers, structures, wiring/protection, accessories and services where relevant.
- Search, category/brand filters, sort, detail view, create/edit/archive and stock visibility. Prices and stock must come from stored data.

## C. Documents and generated records

- Attach bills, site photos, signed proposals, invoices, installation checklists, warranty docs, completion reports and customer/site records.
- Store filename/type/date/uploader metadata, secure access rules and downloadable links; restrict sensitive documents by role.
- Generate PDF quotes/proposals/reports with unique references and predictable filenames.

# 6. Orders, project execution, My Orders and inventory

## A. Quotation-to-order/project conversion

- An accepted quotation creates a linked customer order/project without retyping core customer, line-item, amount or site details. Preserve the source quotation and display the relationship on both records.
- Order fields: order ID/date, quote reference, customer/site, project type/capacity, contract value, confirmed scope, payment schedule, delivery target, documents, status, cancellation reason and owner.
- Project fields: project ID, project manager, assigned sales/engineering/installation team, planned/actual dates, priority, milestone, percent complete, risks/blockers, materials required/allocated, and customer update history.
- Suggested milestones: Order Confirmed -> Documents Pending -> Site/Technical Approval -> Material Planning -> Material Ready -> Installation Scheduled -> Installation In Progress -> Inspection/Testing -> Commissioned/Handed Over -> Closed. Support hold/exception states with a recorded reason.
- Installation record: crew, schedule, checklist, serial numbers where tracked, install photos, safety notes, testing/commissioning readings, customer sign-off, completion report and handover documents.
- Show timeline, milestone timestamps, stuck/overdue projects, completion reports and progress updates.

## B. My Orders

- Separate Pending, Active, Completed and Cancelled views; search by order/customer/quote, status/date filters, order detail, timeline, linked quote/project and related document downloads.
- If customer portal access exists, customers must only view their own projects. Otherwise make the page internal-only and document that boundary.

## C. Inventory, materials and procurement

- Track available, reserved, allocated, issued, returned and damaged quantities. Deduct/reserve stock only through explicit supported transactions; prevent negative stock unless a protected admin setting deliberately permits it.
- Link bill of materials/material requirements to projects/orders; show required vs available vs allocated quantities and shortage warnings.
- Stock movement log: date, warehouse/branch, reference, quantity, user and reason. Include supplier/vendor name, contact, address, tax identifier if applicable, status and notes.
- Low-stock alerts, material/product/order/project reports and reconciliation to stock transactions. Purchase requests/orders and goods-received workflow are bonus items.

# 7. Service tickets, reports and administration

## A. Tickets, complaints and AMC

- Ticket ID, customer, linked project/order/product, category, issue, priority/source, opened date, assignee/team, status, target resolution date, resolution notes, parts/costs, feedback and timestamps.
- Ticket states: Open, Assigned, In Progress, Waiting for Customer/Parts, Resolved, Closed. Support assignment, comments, status history, search/filter, resolution report and closure.
- AMC record: contract number, customer/site, start/end dates, coverage, visit frequency, SLA, amount, next service date, status and renewal reminder. A basic record is acceptable if recurring automation is not completed.

## B. Reports and exports

| Report | Minimum coverage |
|---|---|
| **Lead** | ID, source, owner, stage, creation date, follow-up, expected value and win/loss; date/source/status filters. |
| **Customer** | Profile, city/state, related leads, visits, quotes, projects/orders and tickets. |
| **Quotation** | Quote number, customer, date, subtotal/tax/total, status, owner, validity and conversion. |
| **Sales/revenue** | Period, team member, won deals, conversion and recorded revenue/payment information. Do not infer revenue without financial records. |
| **Order/project** | Status, system type/capacity, manager, milestone, planned vs actual dates, delays and completion. |
| **Product/tax** | Stock/low-stock and tax summary derived from stored quote/invoice records and configured rates. |
| **Ticket/AMC** | Open/overdue/resolved tickets, time-to-resolution if available, coverage and upcoming renewals. |
| **Monthly business** | Leads, quote conversion, orders/projects, sales value, recorded payments/outstanding and service workload from real data. |

Support date-range/status/owner filters, sorting, totals and page size. Provide CSV/Excel-compatible exports and PDF reports. Exported rows/totals must match the current filtered view.

## C. Admin panel and security

- User management: name, email/login, phone, role, branch/team, active/inactive, and last login if available; create/invite/reset/deactivate users.
- Role/permission management: restrict view/create/edit/delete/export for sensitive modules. Demonstrate at least two roles with different server-enforced permissions.
- Audit logs for important actions: user, action, entity/reference, timestamp and change summary. Never log passwords or raw secrets.
- Backup/export plan and REST API approach. Prefer one authenticated sample endpoint. Full automated restore can be documented if outside the time box.
- Security: server-side checks, secure password hashing, validation, safe uploads, secrets in environment variables/secure server storage, and no public access to private records.

# 8. Mandatory dynamic Settings tab - all applicable settings

**NON-NEGOTIABLE REQUIREMENT**

**Create one dedicated Settings tab/page. All business configuration listed below must be editable by authorized admins, persist in the database, and take effect across the app without changing source code. Do not leave settings as hard-coded constants or decorative controls.**

Settings should be organized into clear sub-tabs/cards with Save, Cancel, validation, confirmation for sensitive changes, success/error messages, permissions and an audit trail. Settings that are not implemented must be clearly marked as unavailable rather than shown as working.

## A. Application branding and organization profile

- Change displayed application/website name, organization/company name, logo, favicon, login-page logo, sidebar/header branding, footer text and theme/accent color if supported.
- Company details: legal name, display name, address, phone, support email, website URL, GSTIN/tax identifiers, invoice address and other contact details used in PDFs.
- Logo/name changes must appear in navigation, login screen where applicable, quotation PDFs, report headers and generated documents. Clarify that this changes app branding/display name, not the DNS domain registration itself.
- Allow secure logo upload with file type/size validation, preview, replace and remove/reset-to-default.

## B. Account, password and security settings

- Allow a signed-in user to change their password using current password, new password and confirmation. Show policy requirements, mismatch/weak-password errors and success confirmation.
- Admin-managed password reset/invitation flow, user activation/deactivation, session expiry/idle timeout and password policy where feasible.
- Never display or recover existing plaintext passwords. Hash passwords securely, do not log credentials, and do not put secrets in browser code or Git.
- Optional if time permits: two-factor authentication, login lockout, session/device list and security notification settings.

## C. Users, roles, permissions and branches

- Configure roles, module access and actions (view/create/edit/delete/export/approve/settings). Protect settings pages and endpoints by permission, not only by hiding navigation items.
- Configure branch/warehouse list, territory/region, assignment rules, team names and access scope.
- Configure company/fiscal-year details, numbering sequences and prefixes for leads, quotes, orders, invoices, projects, tickets and AMC contracts.

## D. Dynamic fields, lists and workflow configuration

- Add/edit/deactivate lead sources, lead stages, customer types, property types, product categories, units, priorities, ticket types/statuses, loss reasons, task types, document types and notification categories.
- Where feasible, allow admins to define custom fields (text, number, date, dropdown, checkbox), field label, required flag, default value, validation, visibility and module placement.
- Configure quotation statuses, project milestones, ticket workflow, AMC statuses and allowed transitions. New configured values must appear in forms, filters and reports.
- Configure table columns and defaults where feasible. Do not allow invalid configurations to break existing records; preserve old values or provide migration/deactivation behavior.

# 9. Dynamic Settings tab - continued

## E. Solar calculator and commercial assumptions

- Hybrid calculator defaults: peak sun hours, performance ratio/system losses, panel wattage, battery chemistry, usable depth of discharge, battery/inverter efficiency, design margin, voltage options, export assumptions and currency/unit preferences.
- Rates and assumptions: electricity tariffs, export/net-metering prices, O&M defaults, component prices, installation/transport cost defaults, tax/GST rates, discount limits and optional subsidy rules.
- Keep each configuration with value, unit, effective date, status, source/reference and updated-by metadata. For real policy assumptions, capture applicable state/utility and source/date; otherwise label values as Sample/Demo.
- Changes to calculator settings must affect new calculations and reports. Previously saved quotes/calculations should retain a snapshot of assumptions or clearly display which version was used.

## F. Product, inventory and procurement settings

- Product category, brand, warranty defaults, unit list, tax rate, price defaults, serial/batch tracking toggle, reorder threshold, warehouse/branch, stock reservation behavior and negative-stock policy.
- Configure allowed product types and component attributes for panels, hybrid inverters, batteries, pumps, EV chargers, structures and accessories.

## G. Documents, numbering and templates

- Configure quote/invoice/order/project/ticket/AMC numbering prefixes, sequence behavior, validity days, payment terms, warranty/maintenance text, default notes and terms.
- Editable branded templates for quotation PDF, hybrid calculator report, site-visit report, installation/commissioning report, invoice/receipt, ticket resolution report and customer communications.
- Enable/disable QR code, signature block, footer/contact block, logo display and PDF/CSV/Excel-compatible export options where supported.

## H. WhatsApp Business and communication settings

- Provide a WhatsApp integration settings page with provider/mode (Demo or configured provider), business phone/sender number, business/account identifiers and approved template names. Use the provider/API version required by the selected integration.
- Support secure server-side credential configuration, test-connection action, connection status, webhook/callback configuration if implemented and a clear message when API credentials are missing. Never expose permanent access tokens in front-end code or ordinary logs.
- Editable message templates with safe placeholders, preview and validation for lead assignment, follow-up reminder, site visit confirmation, quotation ready/shared, quote accepted, project milestone update, invoice/payment update, ticket status and AMC renewal.
- Add WhatsApp actions where useful: share quotation PDF/link, share calculator/report link, send approved template message and open a customer conversation. If provider/API is not configured, run a clearly labelled demo mode and do not claim delivery.
- Store message history: linked record, recipient, template, time, initiating user, status (queued/sent/delivered/failed when provider supports status), provider response/error and retry action where appropriate.
- Respect customer consent/opt-in and opt-out state. Validate phone numbers with country code; provide safe error feedback for invalid numbers, missing variables or API errors.

## I. Email, notifications and reminders

- Configurable SMTP/provider mode and sender display name/address; use server-side secret storage. Include a test email action only if implemented.
- Editable email templates for quotes, visits, project updates, ticket updates, invoices and AMC reminders. Optional SMS/push channels may be documented if not implemented.
- Configure reminders, lead follow-up intervals, task due reminders, notification channels, working hours and ticket/project SLA settings.

# 10. Dynamic Settings validation and feature acceptance

## J. Regional, finance, integration and system settings

- Locale, timezone, date/time format, currency symbol/precision, measurement units, rounding behavior and default country calling code.
- Tax rules/GST rates, company fiscal year and financial report parameters must be editable by authorized users, validated and audit logged.
- REST API configuration: enabled/disabled state, API client/key metadata and rotation/revocation workflow if implemented. Never reveal a complete secret after saving it.
- Map/GPS, signature, payment, backup and external-data integration status should show Configured / Not configured / Demo. Secrets belong server-side.
- Configure data-export formats, backup reminder/schedule settings, retention preferences and audit-log filter settings where feasible. Dangerous restore/delete actions need confirmation and authorization.

## Settings acceptance tests - mandatory

| ID | Test | Expected result |
|---|---|---|
| **SET-01** | Change app/website display name and logo | Branding changes in navigation/login where used and in generated quotation/report PDFs; a page refresh does not revert it. |
| **SET-02** | Change own password | Current password is validated; new/confirm mismatch is rejected; new password works and old one fails. |
| **SET-03** | Change dropdown/workflow value | New lead source or stage appears in forms and filters; a changed project/ticket workflow is reflected in the appropriate screens. |
| **SET-04** | Change hybrid defaults | Changing peak sun hours, battery DoD, loss factor or panel wattage changes relevant results and shows the selected assumptions. |
| **SET-05** | Configure WhatsApp in Demo mode | Template preview works, demo action is visibly marked simulated, and the system does not falsely claim sent/delivered status. |
| **SET-06** | Missing WhatsApp credentials | The app displays a useful setup message and never exposes or requests credentials in a public/client-side field. |
| **SET-07** | Role restrictions | A non-admin cannot save protected settings by UI or direct HTTP/API request. |
| **SET-08** | Audit history | Important settings changes record user, timestamp, setting category and safe change summary; secrets/password values are not logged. |
| **SET-09** | Persistence | Saved settings remain after logout/login and server restart using database-backed configuration. |

## Other required or visible features

- Responsive desktop/tablet/mobile design, accessible labels, search/filter/sort/pagination and sensible error states.
- PDF and CSV/Excel-compatible exports, quote QR code, signature capture/display area, document management, GPS/site photos, notifications, audit trail and REST API integration point or documented API design.
- Email/WhatsApp live delivery, live maps, external subsidy APIs, electronic signatures, payment gateway and production backups may be simulated only when labelled clearly and described in README.

# 11. Definition of done - end-to-end acceptance checklist

| ID | Acceptance test |
|---|---|
| **AT-01** | Log in with demo credentials; demonstrate at least two roles with different server-enforced permissions. |
| **AT-02** | Create a lead, assign it, add follow-up/customer note, and find it using search/filter. |
| **AT-03** | Schedule and complete a site visit with survey fields, photo/document support and visit history/report. |
| **AT-04** | Run hybrid calculator; show changed inputs, units, assumptions, formula/intermediate values and changing outputs; export a report. |
| **AT-05** | Run solar investment and EV income calculators; verify input changes affect results logically. |
| **AT-06** | Create itemized hybrid and solar water-pump quotations, calculate configured taxes/totals, generate PDF and change quote status. |
| **AT-07** | Accept a quotation and convert it into a linked order/project without losing key data; update milestones and show progress. |
| **AT-08** | View catalogue/stock, link required materials to a project and show stock shortage/low-stock information. |
| **AT-09** | Create, assign, update and close a ticket linked to customer/project; show AMC record or renewal workflow. |
| **AT-10** | Dashboard KPIs and core reports reflect stored records; export data and verify filters/totals. |
| **AT-11** | Demonstrate dynamic settings: app name/logo, password change, a new dropdown/workflow setting, calculator assumptions and WhatsApp Demo/configured mode. |
| **AT-12** | README setup works; seed data/demo login exist; no secrets committed; known limitations are listed. |
| **AT-13** | Submit full-project recording and Claude-only evidence within the 30-hour limit. |

## Important technical expectations

- Related screens must update consistently from one stored source of truth. Avoid duplicate copies of the same business setting across unrelated forms.
- Show loading/success/failure feedback, validate user input on both client and server where relevant, and handle empty states.
- For integrations, expose real status only when a real provider/API response supports it. A local “success” toast is not proof that WhatsApp/email was delivered.
- Store a snapshot of quote/calculation values so editing current settings does not silently alter previously accepted commercial documents.

# 12. Submission format and next-round interview

## Submit these items together

| Deliverable | What to include |
|---|---|
| **1. Application** | Working URL or local run steps, demo credentials, seed data and core flows accessible to evaluator. |
| **2. Source** | Repository/archive, database migrations/schema, environment sample with no secrets and dependency/lock files. |
| **3. README** | Setup, stack, architecture, modules, database, demo accounts/roles, tests, deployment notes, limitations and future work. |
| **4. Screen recording** | One complete walkthrough, preferably 12-20 minutes. Show the live running application, not only slides/screenshots; narrate or explain the key steps. |
| **5. Claude evidence** | Prompt log/export or screenshots showing Claude AI-only coding; explain how generated code was verified. |
| **6. Status sheet** | Mark every feature Fully working / Simulated / Partial / Not implemented and list bugs/dependencies. |

## Next-round interview will be based on your project

- Explain architecture, data model/relationships, authentication, roles and permission enforcement.
- Walk through lead -> survey -> hybrid calculation -> quote -> order/project -> materials -> installation -> ticket/service. Explain how linked records are saved and validated.
- Explain hybrid sizing formula assumptions and units. The interviewer may change appliance usage, backup target, PV generation, battery depth of discharge or costs and ask you to predict the result.
- Explain why you chose your architecture, how you prompted Claude, how you identified incorrect generated code, and how you debugged and tested it.
- Demonstrate dynamic settings and explain how changes propagate to forms, calculations, documents and reports without manual code changes.
- Explain WhatsApp configuration, approved templates, consent/opt-out, demo vs real sending, delivery-status limitations and secret handling.
- Answer questions about validation, error handling, security, REST APIs, indexing, performance, auditability, backups, deployment and known limitations.
- Be ready to make a small change live using Claude AI, explain the diff and show the test result.

## Suggested evaluation rubric

| Area | Weight | What is evaluated |
|---|---|---|
| **Core workflow functionality** | 25% | Connected modules and reliable end-to-end data flow. |
| **Solar domain logic** | 20% | Hybrid calculator inputs/results, assumptions, cost/tax estimates, inventory/project logic. |
| **Dynamic settings and integrations** | 15% | Settings persistence, branding/password/workflow changes, WhatsApp/email demo or real integration behavior. |
| **Code and data design** | 15% | Structure, relationships, validation, error handling, documentation. |
| **UI/UX and usability** | 10% | Responsive screens, forms, search/filter, feedback and navigation. |
| **Security and permissions** | 10% | Auth, server permissions, secret handling and safe documents. |
| **Recording / AI evidence** | 5% | Clear walkthrough, Claude-only evidence and ability to explain decisions. |

# 13. Research references and assumptions

This assignment retains the feature scope in the supplied Solar CRM Management System proposal and revises the calculator scope as requested: separate off-grid and on-grid calculator specifications are replaced by one hybrid solar calculator. The solar investment/income calculator and EV income calculator remain in scope.

Additional product workflow ideas for planning are referenced below. They are for feature research only; the candidate must create an original implementation and must not copy proprietary code or branding.

## Public feature references

**Solar CRM - Features**
https://getsolarcrm.com/features

Lead pipeline, contacts, tasks/calendar, tax-aware quotes, site visits, installation, inventory, projects, support and permissions.

**SOLARCRM - Features**
https://www.solarcrm.co/features

Lead progress, quote/proposal processes, project tracking and reporting.

**SolarNeo - Features**
https://solarneo.in/features

Survey-first mobile workflows, quotation/subsidy estimates, operations, inventory and role-based views.

**Solar CRM for India**
https://solarcrm.in/

Indian installer workflows, GST quotations, configurable subsidy rules, installation milestones, AMC and tickets.

**Google Play - Solar CRM**
https://play.google.com/store/apps/details?id=in.youwegroup.solarcrm

Lead stages, follow-ups, solar calculations, quotations/invoicing and site-visit documents.

## Important assumptions

- Subsidy, net metering, tariffs, GST treatment, system sizing and financial outputs depend on jurisdiction, utility, product specifications and project circumstances.
- Sample values must be labeled Demo/Sample. Real policy values must include source, applicable state/utility and effective date, and remain configurable.
- Changing global settings should not silently rewrite historical quotes or reports already issued to customers; preserve a snapshot/version or show the version used.
- A complete working core flow is more valuable than many decorative screens. Disclose all unfinished or simulated features honestly.

**END OF FIRST-ROUND ASSIGNMENT**